Advanced Topics in Communication Systems

Lecture Notes

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Contents

Notations

- x : Scalar
- $\mathbf{x}$: Vector
- $\mathbf{X}$: Matrix
- $\mathbb{E}[\cdot]$: Expectation Operator
- $\mathrm{Var}(\cdot)$: Variance Operator
- $\mathrm{Cov}(\cdot,\cdot)$: Covariance Operator
- $\mathbb{R}$: Real
- $\mathbb{C}$: Complex
- $f_Y(y)$: PDF - Probability Density Function
- $F_Y(y)$: CDF - Cumulative Density Function
- $Q(\cdot)$: Q-function

Gaussian or Normal Random Variables

A random variable (RV) X is called a Gaussian or Normal random variable if its probability density function (PDF) is given by:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in \mathbb{R},\ \mu \in \mathbb{R},\ \text{and } \sigma > 0 \tag{1.1}$$

$X \sim \mathcal{N}\left(\mu, \sigma^2\right)$: Symbolic notation that X is Gaussian or normal RV with parameters μ, σ^2.

A RV Y distributed as $\mathcal{N}(0,1)$ is referred to as a standard normal RV.

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}, \quad -\infty < y < \infty \tag{1.2}$$

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dy = 1 \tag{1.3}$$

$$\mu_Y = \mathbb{E}[Y] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-\frac{y^2}{2}} dy = 0 \tag{1.4}$$

$$\begin{aligned} \mathbb{E}\left[Y^2\right] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-\frac{y^2}{2}} dy \\ &= \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} y^2 e^{-\frac{y^2}{2}} dy \end{aligned}$$

put $v = \frac{y^2}{2} \Rightarrow dv = ydy$

$$\begin{aligned} \mathbb{E}\left[Y^2\right] &= \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} \sqrt{2v} e^{-v} dv \\ &= \frac{2}{\sqrt{\pi}} \int_{0}^{\infty} v^{\frac{1}{2}} e^{-v} dv \end{aligned} \tag{1.5}$$

Gamma Function

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad \Re(z) > 0 \tag{1.6}$$

$$\Gamma(1) = \int_0^\infty e^{-t} dt = 1$$

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}} e^{-t} dt$$

put $t = \frac{v^2}{2} \Rightarrow dt = vdv$

$$\begin{aligned}\therefore \quad \Gamma\left(\frac{1}{2}\right) &= \int_0^\infty \frac{\sqrt{2}}{v} e^{-\frac{v^2}{2}} vdv \\ &= \sqrt{2}\int_0^\infty e^{-\frac{v^2}{2}} dv \\ &= 2\sqrt{\pi}\int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv \\ \Gamma\left(\frac{1}{2}\right) &= \sqrt{\pi}\end{aligned} \tag{1.7}$$

$$\Gamma(z+1) = z\Gamma(z) \tag{1.8}$$

For positive integer k,

$$\begin{aligned}\Gamma(k+1) &= k\Gamma(k) \\ &= k(k-2)\Gamma(k-2) \\ &= k(k-1)(k-2)\cdots 2 \cdot 1 \cdot \Gamma(1) \\ &= k!\end{aligned} \tag{1.9}$$

Using eqns. (1.7) and (1.8) in eqn. (1.5),

$$\begin{aligned}\therefore \quad \mathbb{E}\left[Y^2\right] &= \frac{2}{\sqrt{\pi}}\int_0^\infty v^{\frac{1}{2}} e^{-v} dv \\ &= \frac{2}{\sqrt{\pi}}\int_0^\infty v^{\frac{3}{2}-1} e^{-v} dv \\ &= \frac{2}{\sqrt{\pi}}\Gamma\left(\frac{3}{2}\right) = \frac{2}{\sqrt{\pi}}\frac{1}{2}\Gamma\left(\frac{1}{2}\right) \\ &= 1\end{aligned} \tag{1.10}$$

CDF of the Standard Normal RV

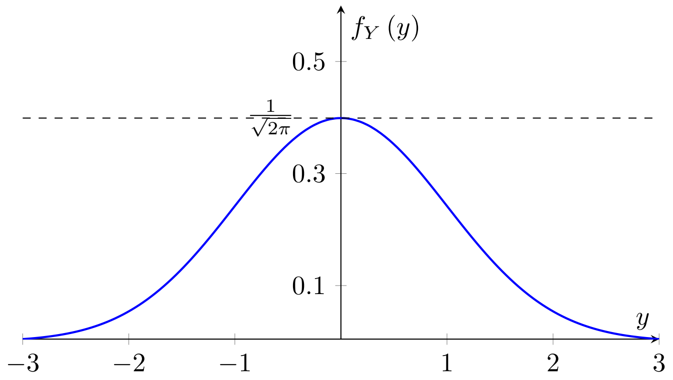
$$\begin{aligned}F_Y(y) &= \Pr(Y \le y) \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^y e^{-\frac{v^2}{2}} dv \\ &= 1 - \frac{1}{\sqrt{2\pi}}\int_y^\infty e^{-\frac{v^2}{2}} dv \\ F_Y(y) &= 1 - Q(y)\end{aligned} \tag{1.11}$$

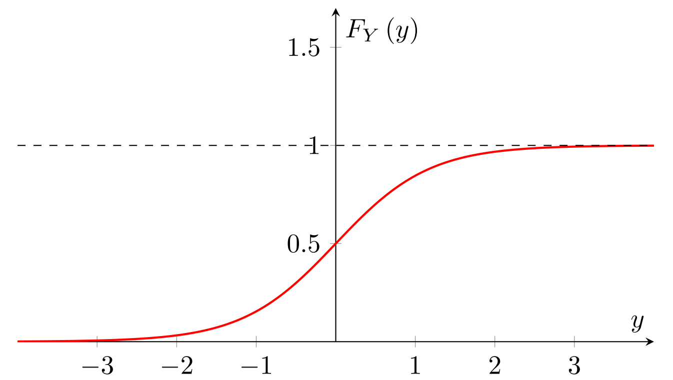
Gaussian Q-Function

$$Q(y) \triangleq \frac{1}{\sqrt{2\pi}}\int_y^\infty e^{-\frac{v^2}{2}} dv \tag{1.12}$$

Properties

- $Q(-y) = 1 - Q(y)$
- $Q(-\infty) = 1$
- $Q(0) = \frac{1}{2}$
- $Q(\infty) = 0$





To find the PDF of $aY + b$ when $Y \sim \mathcal{N}(0,1)$

Let $V = aY + B,$ where a & b are constants $\in \mathbb{R}$.

$$\begin{aligned} F_V(v) &= \Pr(V \le v) \\ &= \Pr(aY + b \le v) \\ &= \Pr(aY \le v - b) \\ &= \Pr\left(Y \le \frac{v-b}{a}\right) \quad \text{if } a > 0 \\ &= \Pr\left(Y \ge \frac{v-b}{a}\right) \quad \text{if } a < 0 \end{aligned} \tag{1.13}$$

If $a > 0$,

$$F_V(v) = F_Y\left(\frac{v-b}{a}\right)$$

Differentiating both sides wrt v, we get,

$$f_V(v) = \frac{d\left(\frac{v-b}{a}\right)}{dv} \cdot \frac{F_Y\left(\frac{v-b}{a}\right)}{d\left(\frac{v-b}{a}\right)}$$

$$f_V(v) = \frac{1}{a} f_Y\left(\frac{v-b}{a}\right) \tag{1.14}$$

If $a < 0$,

$$F_V(v) = \Pr\left(Y \geq \frac{v-b}{a}\right)$$
$$= 1 - \Pr\left(Y \geq \frac{v-b}{a}\right) = 1 - F_Y\left(\frac{v-b}{a}\right)$$

(since there are no impulses in $f_Y(\cdot)$ and hence there are no discontinuities in $F_Y(\cdot)$)
Differentiating both sides wrt v, we get,

$$f_V(v) = -\frac{1}{a} f_Y\left(\frac{v-b}{a}\right) \tag{1.15}$$

Combining eqns. (1.14) and (1.15), we get,

$$f_V(v) = \frac{1}{|a|} f_Y\left(\frac{v-b}{a}\right) \tag{1.16}$$

$$\begin{aligned} \Rightarrow \quad f_V(v) &= \frac{1}{\sqrt{2\pi}|a|} e^{-\frac{1}{2}\left(\frac{v-b}{|a|}\right)^2}, \quad v \in \mathbb{R} \\ \text{or} \quad f_V(v) &= \frac{1}{\sqrt{2\pi a^2}} e^{-\frac{(v-b)^2}{2a^2}}, \quad v \in \mathbb{R} \end{aligned} \tag{1.17}$$

Mean μ_V,

$$\mu_V = \mathbb{E}[aY + b] = a \cdot \mathbb{E}[Y + b] = b \tag{1.18}$$

Variance $\mathrm{Var}(V)$,

$$\begin{aligned} \mathrm{Var}(V) &= \mathbb{E}\left[(V - \mu_V)^2\right] \\ &= \mathbb{E}\left[(aY + b - b)^2\right] \\ &= a^2 \mathbb{E}\left[Y^2\right] = a^2 \end{aligned} \tag{1.19}$$

$$\therefore \quad V \sim \mathcal{N}\left(b, a^2\right), \quad Y = \frac{V-b}{a} \sim \mathcal{N}(0,1)$$

Note

If $Y \sim \mathcal{N}\left(\mu, \sigma^2\right)$, then $V = aY + b \sim \mathcal{N}\left(a\mu + b, a^2\sigma^2\right)$.

Moments

If $X \sim \mathcal{N}\left(\mu, \sigma^2\right)$, then $Y = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)$.

$$\mathbb{E}\left[Y^k\right] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \underbrace{y^k e^{-\frac{y^2}{2}}}_{\substack{\text{odd function of } y \text{ for } k = 1,3,5,\dots \\ \text{even function of } y \text{ for } k = 2,4,6,\dots}} dy \tag{1.20}$$

$$\mathbb{E}\left[Y^k\right] = 0 \quad \text{for } k = 1,3,5,\dots \tag{1.21}$$

For $k = 2l,\ l = 1,2,3,\dots$

$$\begin{aligned} \mathbb{E}\left[Y^k\right] &= \mathbb{E}\left[Y^{2l}\right] \\ &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} y^{2l} e^{-\frac{y^2}{2}} dy \end{aligned}$$

put $v = \frac{y^2}{2} \Rightarrow dv = ydy$,

$$\begin{aligned} \mathbb{E}\left[Y^{2l}\right] &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} (\sqrt{2v})^{2l} e^{-v} y^{-1} dv \\ &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} (\sqrt{2v})^{2l-1} e^{-v} dv \\ &= \frac{2^{l+\frac{1}{2}}}{\sqrt{2\pi}} \int_0^{\infty} v^{l-\frac{1}{2}} e^{-v} dv \end{aligned}$$

Using eqns. (1.6) and (1.8)

$$\mathbb{E}\left[Y^{2l}\right] = \frac{2^l}{\sqrt{\pi}}\Gamma\left(l + \frac{1}{2}\right), \quad l = 1, 2, 3, \ldots \tag{1.22}$$

Therefore,

$$\mathbb{E}\left[(X-\mu)^k\right] = \begin{cases} 0 & k = 1, 3, 5, \ldots \\ \frac{2^{\frac{k}{2}}}{\sqrt{\pi}}\Gamma\left(\frac{k+1}{2}\right)\sigma^k & k = 2, 4, 6, \ldots \end{cases} \tag{1.23}$$

Advanced Topics in Communication Systems	Thurs 08 Jan 2026
Lecture 2	

Types of Random Variables

Scalar $\mathbb{R}$, Vector $\mathbb{R}$, Scalar $\mathbb{C}$ and Vector $\mathbb{C}$

Characteristic Function

It is sometimes easier to characterise a random variable by its CF rather than by its PDF. Also called moment generating function (MGF).

$$\begin{aligned}\varphi_X(j\omega) &= \mathbb{E}\left[e^{j\omega X}\right] \\ &= \mathbb{E}\left[\sum_{k=0}^{\infty}\frac{(j\omega X)^k}{k!}\right] \\ &= \sum_{k=0}^{\infty}\frac{(j\omega X)^k}{k!}\mathbb{E}\left[X^k\right] \\ \varphi_X(j\omega) &= 1 + j\omega\mathbb{E}[X] + \frac{(j\omega)^2}{2}\mathbb{E}\left[X^2\right] + \cdots\end{aligned} \tag{2.1}$$

To find $\mathbb{E}[X]$,

$$\begin{aligned}\left.\frac{d\varphi_X(j\omega)}{\omega}\right|_{\omega=0} &= 0 + j\mathbb{E}[X] + 0 + \cdots \\ \left.\frac{1}{j}\frac{d\varphi_X(j\omega)}{d\omega}\right|_{\omega=0} &= \mathbb{E}[X]\end{aligned} \tag{2.2}$$

Similarly for $\mathbb{E}\left[X^2\right]$,

$$\begin{aligned}\left.\frac{d^2\varphi_X(j\omega)}{\omega^2}\right|_{\omega=0} &= 0 + 0 + j^2\mathbb{E}\left[X^2\right] + 0 + \cdots \\ \left.\frac{1}{j^2}\frac{d^2\varphi_X(j\omega)}{d\omega^2}\right|_{\omega=0} &= \mathbb{E}\left[X^2\right]\end{aligned}$$

In general,

$$\left.\frac{1}{j^k}\frac{d^k\varphi_X(j\omega)}{d\omega^k}\right|_{\omega=0} = \mathbb{E}\left[X^k\right] \tag{2.3}$$

Gaussian Characteristic Function

$$\begin{aligned}\varphi_X(j\omega) &= \mathbb{E}\left[e^{j\omega X}\right], \quad \text{where } X \sim \mathcal{N}\left(\mu, \sigma^2\right) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}}\int_{-\infty}^{\infty} e^{j\omega x}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx\end{aligned} \tag{2.4}$$

Take

$$\begin{aligned}j\omega x - \frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2 &= -\frac{1}{2}\left(\frac{x^2+\mu^2-2\mu x}{\sigma^2} - 2j\omega x\right) \\ &= -\frac{1}{2\sigma^2}\left(x^2+\mu^2-2\mu x-2j\omega x\sigma^2\right) \\ &= -\frac{1}{2\sigma^2}\left[x^2+\mu^2-2\left(\mu+j\omega\sigma^2\right)x+\mu^2\right] \\ &= -\frac{1}{2\sigma^2}\left[x^2+\mu^2-2\left(\mu+j\omega\sigma^2\right)x+\left(\mu+j\omega\sigma^2\right)^2+\mu^2-\left(\mu+j\omega\sigma^2\right)^2\right]\end{aligned}$$

$$
\begin{aligned}
&= -\frac{1}{2\sigma^2}\left[x-\left(\mu+j\omega\sigma^2\right)\right]^2 - \frac{1}{2\sigma^2}\left[\mu^2-\left(\mu+j\omega\sigma^2\right)\right] \\
&= -\frac{1}{2\sigma^2}\left[x-\left(\mu+j\omega\sigma^2\right)\right]^2 - \frac{1}{2\sigma^2}\left(\mu^2-\mu^2-2j\omega\mu\sigma^2+\omega^2\sigma^4\right) \\
&= -\frac{1}{2}\left[\frac{x-\left(\mu+j\omega\sigma^2\right)}{\sigma}\right]^2 + \frac{1}{2}\left(2j\omega\mu-\omega^2\sigma^2\right)
\end{aligned}
$$

So,

$$
\varphi_X(j\omega) = e^{\frac{1}{2}\left(2j\omega\mu-\omega^2\sigma^2\right)} \cdot \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[\frac{x-\left(\mu+j\omega\sigma^2\right)}{\sigma}\right]^2} dx}_{=1} \tag{2.5}
$$

$$
\therefore \quad \varphi_X(j\omega) = e^{j\omega\mu-\frac{1}{2}\omega^2\sigma^2}, \quad \text{when } X \sim \mathcal{N}\left(\mu,\sigma^2\right)
$$

For the standard normal RV $Y,\ \mu = 0$ and $\sigma^2 = 1$,

$$
\therefore \quad \varphi_Y(j\omega) = e^{-\frac{1}{2}\omega^2} \tag{2.6}
$$

Note

- $X \sim \mathcal{N}\left(\mu,\sigma^2\right),\ Y \sim \mathcal{N}(0,1)$

$$
\Rightarrow Y = \frac{X-\mu}{\sigma} \quad \Rightarrow X = \sigma Y + \mu
$$

$$
\begin{aligned}
\varphi_X(j\omega) &= \mathbb{E}\left[e^{j\omega X}\right] = \mathbb{E}\left[e^{j\omega(\sigma Y+\mu)}\right] \\
&= e^{j\omega\mu}\mathbb{E}\left[e^{j\omega\sigma Y}\right] \\
&= e^{j\omega\mu}\varphi_Y(j\omega\sigma) \\
&= e^{j\omega\mu}e^{-\frac{1}{2}\omega^2\sigma^2} = e^{j\omega\mu-\frac{1}{2}\omega^2\sigma^2}
\end{aligned}
$$

- If $X \sim \mathcal{N}\left(\mu,\sigma^2\right)$ and $\sigma^2 \to 0$, then

$$
\begin{aligned}
\varphi_X(j\omega) &\to e^{j\omega\mu}, \\
f_X(x) &\to \delta(x-\mu)
\end{aligned}
$$

Jointly Gaussian Random Variables: The Multivariate Gaussian Distribution

Random variables $X_1, X_2, \ldots, X_L$ are called jointly Gaussian or jointly normal random variables if their joint PDF is given by,

$$
f_{X_1,X_2,\ldots,X_L}(x_1,x_2,\ldots,x_L) = \frac{1}{(2\pi)^{\frac{L}{2}}\,|\mathbf{K}|^{\frac{1}{2}}}\exp\left\{-\frac{1}{2}\sum_{k=1}^{L}\sum_{l=1}^{L}(x_k-\mu_k)\left(\mathbf{K}^{-1}\right)_{kl}(x_l-\mu_l)\right\},
$$
$$
-\infty < x_1,x_2,\ldots,x_L < \infty \text{ and } -\infty < \mu_1,\mu_2,\ldots,\mu_L < \infty \tag{2.7}
$$

- $\mathbf{K}$ is a real-valued $L \times L$ positive definite matrix.
- $\left(\mathbf{K}^{-1}\right)_{kl}$ denotes the element in the k-th row and l-th column of $\mathbf{K}^{-1}$.

Denoting the random vector $\mathbf{X}$ as,

$$
\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_L \end{bmatrix}
$$

and the vectros $\mathbf{x}$ and $\boldsymbol{\mu}$ as,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_L \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_L \end{bmatrix}$$

We can express the joint PDF of $X_1, X_2, \ldots, X_L$ as the PDF of the random vector $\mathbf{X}$, which is given by,

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{L}{2}} |\mathbf{K}|^{\frac{1}{2}}} \exp\left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{K}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$
$$\mathbf{x} \in \mathbb{R}^L,\ \boldsymbol{\mu} \in \mathbb{R}^L,\ \mathbf{K} \in \mathbb{R}^{L \times L} \text{ and } \mathbf{K} \text{ is positive definite.} \quad (2.8)$$

Note that $\mathbf{K}$ is real-valued and positive definite implies,

- $\det(\mathbf{K}) = |\mathbf{K}| > 0$
- $\mathbf{K}$ is symmetric $\left(\mathbf{K} = \mathbf{K}^\top\right)$
- All eigenvalues of $\mathbf{K}$ are real and positive.

A matrix $\mathbf{M} \in \mathbb{R}^{L\times L}$ is positive definite when for all $\mathbf{x} \in \mathbb{R}^L - \{\mathbf{0}_{L\times 1}\}$, $\mathbf{x}^\top \mathbf{M}\mathbf{x} > 0$.

$$\mathbb{E}[X_k] = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} x_k f_{X_1,\ldots,X_L}(x_1, \ldots, x_L)\, dx_1 \ldots dx_L = \mu_k, \quad k = 1, 2, \ldots, L \quad (2.9)$$

$$\begin{aligned} \mathbb{E}[(X_k - \mu_k)(X_l - \mu_l)] &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} (x_k - \mu_k)(x_l - \mu_l) f_{X_1,\ldots,X_L}(x_1, \ldots, x_L)\, dx_1 \ldots dx_L \\ &= \left(\mathbf{K}^{-1}\right)_{kl}, \end{aligned}$$
$$k = 1, \ldots, L \text{ and } l = 1, \ldots L \quad (2.10)$$

$$\therefore \quad \mathbb{E}[\mathbf{X}] = \boldsymbol{\mu} : \text{ Mean vector} \quad (2.11)$$
$$\mathbb{E}\left[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top\right] = \mathbf{K} : \text{ Covariance matrix} \quad (2.12)$$
$$\mathbb{E}\left[\mathbf{X}\mathbf{X}^\top\right] = \mathbf{K} + \boldsymbol{\mu}\boldsymbol{\mu}^\top = \mathbf{R} : \text{ Correlation matrix} \quad (2.13)$$

We say that $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$
$\mathbf{X}$ is a Gaussian random vector with mean vector $\boldsymbol{\mu}$ and covariance matrix $\mathbf{K}$.

Characteristic function of X or joint characteristic function of $X_1, X_2, \ldots, X_L$

$$\begin{aligned} \varphi_{X_1,\ldots,X_L}(j\omega_1, \ldots, j\omega_L) &= \mathbb{E}\left[e^{j\omega_1 X_1 + \cdots + j\omega_L X_L}\right] \\ \varphi_{\mathbf{X}}(j\boldsymbol{\omega}) &= \mathbb{E}\left[e^{j\boldsymbol{\omega}^\top \mathbf{X}}\right] \\ \varphi_{\mathbf{X}}(j\boldsymbol{\omega}) &= e^{j\boldsymbol{\omega}^\top \boldsymbol{\mu} - \frac{1}{2}\boldsymbol{\omega}^\top \mathbf{K}\boldsymbol{\omega}} \end{aligned} \quad (2.14)$$

We can also write as,

$$\varphi_{X_1,\ldots,X_L}(j\omega_1, \ldots, j\omega_L) = e^{j\sum_{k=1}^{L} \omega_k \mu_k - \frac{1}{2}\sum_{k=1}^{L}\sum_{l=1}^{L} \omega_k (\mathbf{K})_{kl}\omega_l} \quad (2.15)$$

when $L = 1$ we have $\boldsymbol{\mu} = \mu$ and $\mathbf{K} = \sigma^2$

$$\Rightarrow \quad \varphi_X(j\omega) = e^{j\omega\mu - \frac{1}{2}\omega^2\sigma^2} \quad (2.16)$$

From the joint characteristic function, we can easily obtain the marginal distributions of $X_k,\ k = 1, \ldots, L$ putting $\omega_1 = \cdots = \omega_{k-1} = \omega_{k+1} = \cdots = \omega_L = 0$ in $\varphi_{\mathbf{X}}(j\boldsymbol{\omega})$

$$\begin{aligned} \varphi_{X_k}(j\omega_k) &= e^{j\omega_k \mu_K - \frac{1}{2}\omega_k^2 (\mathbf{K})_{kk}} \\ \therefore \quad X_k &\sim \mathcal{N}\left(\mu_k, (\mathbf{K})_{kk}\right), \quad k = 1, \ldots, L \end{aligned} \quad (2.17)$$

If $X_1, \ldots, X_L$ are jointly Gaussian then X_k is marginally Gaussian for $k = 1, \ldots, L$. However, the reverse is not true!

Advanced Topics in Communication Systems Tues 13 Jan 2026

Lecture 3

Whitening

Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$ be an $L \times 1$ Gaussian vector and $\mathbf{V} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where $\mathbf{A} \in \mathbb{R}^{L \times L}$, $\mathbf{b} \in \mathbb{R}^{L}$ and $\det(\mathbf{A}) \neq 0$. Then

$$\mathbb{E}[\mathbf{V}] = \mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b} \tag{3.1}$$

$$\begin{aligned} \operatorname{Var}(\mathbf{V}) &= \mathbb{E}\left[(\mathbf{V} - \boldsymbol{\mu}_{\mathbf{V}})(\mathbf{V} - \boldsymbol{\mu}_{\mathbf{V}})^{\top}\right] \\ &= \mathbb{E}\left[(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})^{\top}\right] \\ &= \mathbb{E}\left[\mathbf{A}(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^{\top}\mathbf{A}^{\top}\right] \\ &= \mathbf{A}\mathbb{E}\left[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^{\top}\right]\mathbf{A}^{\top} \\ &= \mathbf{A}\mathbf{K}\mathbf{A}^{\top} \end{aligned} \tag{3.2}$$

$$\therefore \quad \mathbf{V} \sim \mathcal{N}\left(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\mathbf{K}\mathbf{A}^{\top}\right)$$

Therefore, if $\mathbf{b} = -\mathbf{A}\boldsymbol{\mu}$ and $\mathbf{K} = \left(\mathbf{A}^{\top}\mathbf{A}\right)^{-1}$, then $\mathbf{V} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$ has a $\mathcal{N}(\mathbf{0}, \mathbf{I}_L)$ distribution.

When $\mathbf{V} = \begin{bmatrix} V_1 \\ \vdots \\ V_L \end{bmatrix}$ has a $\mathcal{N}(\mathbf{0}, \mathbf{I}_L)$ distribution we can say that $V_1, \dots, V_L$ are iid $\mathcal{N}(0, 1)$ random variables.

A $\mathcal{N}(\boldsymbol{\mu}, \alpha\mathbf{I})$ Gaussian random vector, where $\alpha > 0$, is called a white random vector.

If $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$ is an $L \times 1$ Gaussian random vector such that $\mathbf{K} = \mathbf{B}\mathbf{B}^{\top}$, where $\mathbf{B}$ is an $L \times L$ real-valued non singular matrix, then $\mathbf{B}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_L)$ and $\mathbf{B}^{-1}\mathbf{X} \sim \mathcal{N}\left(\mathbf{B}^{-1}\boldsymbol{\mu}, \mathbf{I}_L\right)$.

Transforming a Gaussian random vector with correlated elements to another Gaussian random vector with covariance matrix equal to the identity matrix or its scaled version by a linear transformation is called <u>whitening</u>.

Whitening by Eigendecomposition

Let $\lambda_1, \lambda_2, \dots, \lambda_L$ be the eigenvalues of $\mathbf{K}$ (all of them may not be distinct). Since $\mathbf{K}$ is a real-valued symmetric positive definite matrix, $\lambda_1, \dots, \lambda_L$ are real and positive, and it is possible to find a set $\{\mathbf{e}_1, \dots, \mathbf{e}_L\}$ of L linearly independent eigenvectors. $\mathbf{e}_i$ is an eigenvector corresponding to eigenvalue $\lambda_i,\ i = 1, 2, \dots, L$.

Thus,

$$\mathbf{K}\mathbf{e}_i = \lambda_i \mathbf{e}_i, \quad i = 1, \dots, L \tag{3.3}$$

$$\mathbf{K}\begin{bmatrix}\mathbf{e}_i & \dots & \mathbf{e}_L\end{bmatrix} = \begin{bmatrix}\mathbf{e}_i & \dots & \mathbf{e}_L\end{bmatrix}\begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \lambda_L \end{bmatrix} \tag{3.4}$$

Let $\mathbf{E} = \begin{bmatrix}\mathbf{e}_1 & \dots & \mathbf{e}_L\end{bmatrix}$ and $\boldsymbol{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \lambda_L \end{bmatrix} = \operatorname{diag}(\lambda_1, \dots, \lambda_L)$, we get

$$\mathbf{K}\mathbf{E} = \mathbf{E}\boldsymbol{\Lambda} \tag{3.5}$$

Since the columns of $\mathbf{E}$ are linearly independent, $\mathbf{E}$ is full-rank (has rank L), and is therefore invertible (non singular).

$$\therefore \quad \mathbf{K} = \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^{-1} \tag{3.6}$$

Since $\mathbf{K}$ is a symmetric matrix, the linearly independent eigenvectors are also orthogonal.

Note that a real-valued symmetric matrix always has real eigenvalues, and hence real-valued eigenvectors.

In addition, we can normalize the eigenvectors so that $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_L\}$ is an orthonormal set.

$$\Rightarrow \quad \mathbf{e}_i^\top \mathbf{e}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{3.7}$$

$$\begin{aligned} \begin{bmatrix} \mathbf{e}_1^\top \\ \vdots \\ \mathbf{e}_L^\top \end{bmatrix} \begin{bmatrix} \mathbf{e}_1 & \ldots & \mathbf{e}_L \end{bmatrix} &= \mathbf{I}_L \\ \text{i.e.} \quad \mathbf{E}^\top \mathbf{E} &= \mathbf{I}_L \end{aligned} \tag{3.8}$$

Since $\mathbf{E}$ is non-singular, this also implies $\mathbf{E}^\top = \mathbf{E}^{-1}$

$$\Rightarrow \quad \mathbf{E}\mathbf{E}^\top = \sum_{i=1}^{L} \mathbf{e}_i \mathbf{e}_i^\top = \mathbf{I}_L \tag{3.9}$$

$$\therefore \quad \mathbf{K} = \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^{-1} = \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^\top \tag{3.10}$$

Note that $\mathbf{K} = \mathbf{K}^\top$ implies

$$\begin{aligned} \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^{-1} &= \left(\mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^{-1}\right)^\top \\ &= \left(\mathbf{E}^{-1}\right) \boldsymbol{\Lambda}\mathbf{E}^\top \end{aligned}$$

which is consistent with the fact that $\mathbf{E}^\top = \mathbf{E}^{-1}$.

Denoting

$$\boldsymbol{\Lambda}^{\frac{1}{2}} = \operatorname{diag}\left(\lambda_1^{\frac{1}{2}}, \ldots, \lambda_L^{\frac{1}{2}}\right) \tag{3.11}$$

We can write

$$\begin{aligned} \mathbf{K} &= \mathbf{E}\boldsymbol{\Lambda}\mathbf{E}^\top \\ &= \mathbf{E}\boldsymbol{\Lambda}^{\frac{1}{2}}\boldsymbol{\Lambda}^{\frac{1}{2}}\mathbf{E}^\top \\ &= \left(\mathbf{E}\boldsymbol{\Lambda}^{\frac{1}{2}}\right)\left(\mathbf{E}\boldsymbol{\Lambda}^{\frac{1}{2}}\right)^\top \end{aligned} \tag{3.12}$$

Since $\mathbf{K} = \mathbf{B}\mathbf{B}^\top$

$$\therefore \quad \mathbf{B} = \mathbf{E}\boldsymbol{\Lambda}^{\frac{1}{2}} \tag{3.13}$$

and

$$\mathbf{B}^{-1} = \boldsymbol{\Lambda}^{-\frac{1}{2}}\mathbf{E}^{-1} = \boldsymbol{\Lambda}^{-\frac{1}{2}}\mathbf{E}^\top \tag{3.14}$$

Example

$$\mathbf{K} = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

Eignevalues :

$$\begin{aligned} \begin{vmatrix} 3-\lambda & 1 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 1 & 3-\lambda \end{vmatrix} &= 0 \\ (3-\lambda)\left[(3-\lambda)^2 - 1\right] - 1\left[(3-\lambda) - 1\right] + 1\left[1 - (3-\lambda)\right] &= 0 \\ (\lambda-2)(\lambda-2)(\lambda-5) &= 0 \end{aligned}$$

$$\lambda_1 = 2, \ \lambda_2 = 2, \ \lambda_3 = 5$$

Eigenvectors :

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be an eigenvector for eigenvalue 2.

$$\begin{aligned} \mathbf{Kx} = 2\mathbf{x} \quad \Rightarrow \quad 3x_1 + x_2 + x_3 &= 2x_1 && \text{(A)} \\ x_1 + 3x_2 + x_3 &= 2x_2 && \text{(B)} \\ x_1 + x_2 + 3x_3 &= 2x_3 && \text{(C)} \end{aligned}$$

From eqns. (A), (B) and (C) $\quad \Rightarrow \quad x_1 + x_2 + x_3 = 0$

Choose $\mathbf{e}_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$, $\mathbf{e}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. $\{\mathbf{e}_1, \mathbf{e}_2\}$ form an orthonormal set. The choice of $\mathbf{e}_1, \mathbf{e}_2$ is not unique.

Let $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ be an eigenvector for eigenvalue 5.

$$\begin{aligned} \mathbf{Ky} = 2\mathbf{y} \quad \Rightarrow \quad 3y_1 + y_2 + y_3 &= 5y_1 && \text{(D)} \\ y_1 + 3y_2 + y_3 &= 5y_2 && \text{(E)} \\ y_1 + y_2 + 3y_3 &= 5y_3 && \text{(F)} \end{aligned}$$

From eqns. (D), (E) and (F) $\quad \Rightarrow \quad y_1 = y_2 = y_3$

Choose $\mathbf{e}_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

$$\begin{aligned} \therefore \quad \mathbf{E} &= \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \end{bmatrix} \\ \mathbf{E} &= \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ -\frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{bmatrix} \\ \mathbf{E}^{-1} = \mathbf{E}^\top &= \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \\ \mathbf{\Lambda}^{-\frac{1}{2}} &= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{5}} \end{bmatrix} \\ \mathbf{B} &= \mathbf{E}\mathbf{\Lambda}^{\frac{1}{2}} \\ \mathbf{B}^{-1} = \mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{E}^\top &= \begin{bmatrix} \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & -\frac{2}{\sqrt{12}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{15}} & \frac{1}{\sqrt{15}} & \frac{1}{\sqrt{15}} \end{bmatrix} \end{aligned}$$

Therefore, if $\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix} \right)$

$$\mathbf{Y} = \mathbf{B}^{-1}\mathbf{X} = \begin{bmatrix} \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & -\frac{2}{\sqrt{12}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{15}} & \frac{1}{\sqrt{15}} & \frac{1}{\sqrt{15}} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} \dfrac{X_1 + X_2}{\sqrt{12}} - \dfrac{2X_3}{\sqrt{12}} \\ \dfrac{X_1 - X_2}{2} \\ \dfrac{X_1 + X_2 + X_3}{\sqrt{15}} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

Advanced Topics in Communication Systems Thurs 15 Jan 2026

Lecture 4

The Complex Gaussian Random Vector

Let $\mathbf{X}$ and $\mathbf{Y}$ be $L \times 1$ real random vectors, which are jointly Gaussian.
Let $\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}_{\mathbf{X}}$ and $\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}_{\mathbf{Y}}$.

$$\underbrace{\mathbf{K}}_{\text{Covariance matrix}} = \mathbb{E}\left[\begin{bmatrix}\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}} \\ \mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}}\end{bmatrix}\left[(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})^\top \quad (\mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}})^\top\right]\right] = \begin{bmatrix}\mathbf{K}_{\mathbf{XX}} & \mathbf{K}_{\mathbf{XY}} \\ \mathbf{K}_{\mathbf{YX}} & \mathbf{K}_{\mathbf{YY}}\end{bmatrix} \tag{4.1}$$

Thus,

$$\left.\begin{aligned}\mathbf{K}_{\mathbf{XX}} &= \mathbb{E}\left[(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})^\top\right] \\ \mathbf{K}_{\mathbf{YY}} &= \mathbb{E}\left[(\mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}})(\mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}})^\top\right]\end{aligned}\right\} \quad \begin{array}{l}\text{Autocovariance or Covariance} \\ \text{matrix of } \mathbf{X} \text{ or } \mathbf{Y}\end{array} \tag{4.2}$$

$$\left.\begin{aligned}\mathbf{K}_{\mathbf{XY}} &= \mathbb{E}\left[(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})(\mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}})^\top\right] \\ \mathbf{K}_{\mathbf{YX}} = \mathbf{K}_{\mathbf{XY}}^\top &= \mathbb{E}\left[(\mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}})(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})^\top\right]\end{aligned}\right\} \quad \begin{array}{l}\text{Covariance-covariance matrix} \\ \text{of } \mathbf{X} \text{ and } \mathbf{Y}\end{array} \tag{4.3}$$

Note

- $\mathbf{K}$, $\mathbf{K}_{\mathbf{XX}}$ and $\mathbf{K}_{\mathbf{YY}}$ are covariance matrices, hence they are positive definite.
- $\mathbf{K}_{\mathbf{XY}}$ is a cross covariance matrix, and in general, is not positive definite.

$\boldsymbol{\mu}_{\mathbf{X}}, \boldsymbol{\mu}_{\mathbf{Y}} \in \mathbb{R}^L$
$\mathbf{K}_{\mathbf{XX}}, \mathbf{K}_{\mathbf{YY}}, \mathbf{K}_{\mathbf{XY}} \in \mathbb{R}^{L \times L}$
$\mathbf{K} \in \mathbb{R}^{2L \times 2L}$

Joint PDF of X and Y

$$f_{\mathbf{X},\mathbf{Y}}(\mathbf{x}, \mathbf{y}) = \frac{1}{(2\pi)^L |\mathbf{K}|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}\left[(\mathbf{x} - \boldsymbol{\mu}_{\mathbf{X}})^\top \quad (\mathbf{y} - \boldsymbol{\mu}_{\mathbf{Y}})^\top\right] \mathbf{K}^{-1} \begin{bmatrix}\mathbf{x} - \boldsymbol{\mu}_{\mathbf{X}} \\ \mathbf{y} - \boldsymbol{\mu}_{\mathbf{Y}}\end{bmatrix}\right\} \qquad \mathbf{x}, \mathbf{y} \in \mathbb{R}^L \tag{4.4}$$

Let $\mathbf{Z} = \mathbf{X} + j\mathbf{Y}$ denote the $L \times 1$ complex Gaussian random vector.
Thus,

$$\begin{aligned}\mathbf{X} &= \Re(\mathbf{Z}) = \frac{1}{2}\left(\mathbf{Z} + \overline{\mathbf{Z}}\right) \\ \mathbf{Y} &= \Im(\mathbf{Z}) = \frac{1}{2j}\left(\mathbf{Z} - \overline{\mathbf{Z}}\right)\end{aligned} \tag{4.5}$$

$$\mathbb{E}[\mathbf{Z}] = \boldsymbol{\mu}_{\mathbf{Z}} = \boldsymbol{\mu}_{\mathbf{X}} + j\boldsymbol{\mu}_{\mathbf{Y}} \tag{4.6}$$

Let $\mathbf{Z}' = \mathbf{Z} - \boldsymbol{\mu}_{\mathbf{Z}}$, $\mathbf{X}' = \mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}}$ and $\mathbf{Y}' = \mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}}$

$$\left.\begin{aligned}\mathbf{K}_{\mathbf{Z}} &= \mathbb{E}\left[(\mathbf{Z} - \boldsymbol{\mu}_{\mathbf{Z}})(\mathbf{Z} - \boldsymbol{\mu}_{\mathbf{Z}})^{\mathsf{H}}\right] \\ &= \mathbb{E}\left[\mathbf{Z}'\mathbf{Z}'^{\mathsf{H}}\right] \\ &= \mathbb{E}\left[(\mathbf{X}' + j\mathbf{Y}')(\mathbf{X}' - j\mathbf{Y}')^\top\right] \\ &= \mathbb{E}\left[\mathbf{X}'\mathbf{X}'^\top\right] + \mathbb{E}\left[\mathbf{Y}'\mathbf{Y}'^\top\right] + j\left\{\mathbb{E}\left[\mathbf{Y}'\mathbf{X}'^\top\right] - \mathbb{E}\left[\mathbf{X}'\mathbf{Y}'^\top\right]\right\} \\ &= \mathbf{K}_{\mathbf{XX}} + \mathbf{K}_{\mathbf{YY}} + j\left(\mathbf{K}_{\mathbf{XY}}^\top - \mathbf{K}_{\mathbf{XY}}\right)\end{aligned}\right\} \quad \begin{array}{l}\text{Autocovariance or} \\ \text{Covariance matrix of } \mathbf{Z}\end{array} \tag{4.7}$$

$$\left.\begin{aligned}
\tilde{\mathbf{K}}_{\mathbf{Z}} &= \mathbb{E}\left[(\mathbf{Z}-\boldsymbol{\mu}_{\mathbf{Z}})(\mathbf{Z}-\boldsymbol{\mu}_{\mathbf{Z}})^{\top}\right] \\
&= \mathbb{E}\left[\mathbf{Z}'\mathbf{Z}'^{\top}\right] \\
&= \mathbb{E}\left[(\mathbf{X}'+j\mathbf{Y}')(\mathbf{X}'+j\mathbf{Y}')^{\top}\right] \\
&= \mathbb{E}\left[\mathbf{X}'\mathbf{X}'^{\top}\right]-\mathbb{E}\left[\mathbf{Y}'\mathbf{Y}'^{\top}\right]+j\left\{\mathbb{E}\left[\mathbf{Y}'\mathbf{X}'^{\top}\right]+\mathbb{E}\left[\mathbf{X}'\mathbf{Y}'^{\top}\right]\right\} \\
&= \mathbf{K}_{\mathbf{XX}}-\mathbf{K}_{\mathbf{YY}}+j\left(\mathbf{K}_{\mathbf{XY}}^{\top}+\mathbf{K}_{\mathbf{XY}}\right)
\end{aligned}\right\}\ \text{Pseudo-autocovariance or Pseudo-covariance matrix of } \mathbf{Z} \tag{4.8}$$

Thus,

$$\mathbf{K}_{\mathbf{Z}} = \mathbf{K}_{\mathbf{XX}}+\mathbf{K}_{\mathbf{YY}}+j\left(\mathbf{K}_{\mathbf{XY}}^{\top}-\mathbf{K}_{\mathbf{XY}}\right)$$
$$\tilde{\mathbf{K}}_{\mathbf{Z}} = \mathbf{K}_{\mathbf{XX}}-\mathbf{K}_{\mathbf{YY}}+j\left(\mathbf{K}_{\mathbf{XY}}^{\top}+\mathbf{K}_{\mathbf{XY}}\right)$$

$$\Rightarrow \quad \mathbf{K}_{\mathbf{XX}} = \frac{1}{2}\left\{\Re\left(\mathbf{K}_{\mathbf{Z}}\right)+\Re\left(\tilde{\mathbf{K}}_{\mathbf{Z}}\right)\right\} \tag{4.9}$$

$$\mathbf{K}_{\mathbf{YY}} = \frac{1}{2}\left\{\Re\left(\mathbf{K}_{\mathbf{Z}}\right)-\Re\left(\tilde{\mathbf{K}}_{\mathbf{Z}}\right)\right\} \tag{4.10}$$

$$\mathbf{K}_{\mathbf{XY}} = \frac{1}{2}\left\{\Im\left(\tilde{\mathbf{K}}_{\mathbf{Z}}\right)-\Im\left(\mathbf{K}_{\mathbf{Z}}\right)\right\} \tag{4.11}$$

Note that $\mathbf{K}_{\mathbf{Z}}$ is a Hermitian positive definite matrix.
$\mathbf{K}_{\mathbf{Z}}$ is complex-valued and positive definite implies,

- $\det\left(\mathbf{K}_{\mathbf{Z}}\right)=\left|\mathbf{K}_{\mathbf{Z}}\right|>0$
- $\mathbf{K}_{\mathbf{Z}}$ is Hermitian (conjugate symmetric) i.e., $\mathbf{K}_{\mathbf{Z}}=\mathbf{K}_{\mathbf{Z}}^{\mathsf{H}}=\left(\overline{\mathbf{K}}_{\mathbf{Z}}\right)^{\top}$
- All eigenvalues of $\mathbf{K}_{\mathbf{Z}}$ are real and positive.

A matrix $\mathbf{M}\in\mathbb{C}^{L\times L}$ is positive definite when for all $\mathbf{z}\in\mathbb{C}^{L}-\{\mathbf{0}_{L\times 1}\}$, $\mathbf{z}^{\mathsf{H}}\mathbf{M}\mathbf{z}>0$.

$\tilde{\mathbf{K}}_{\mathbf{Z}}$ is a symmetric matrix, in general, it is not positive definite.

If in the joint PDF $f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})$, we substitute

$$\mathbf{X}=\Re(\mathbf{Z}),\ \ \mathbf{Y}=\Im(\mathbf{Z})$$
$$\boldsymbol{\mu}_{\mathbf{X}}=\Re(\boldsymbol{\mu}_{\mathbf{Z}}),\ \ \boldsymbol{\mu}_{\mathbf{Y}}=\Im(\boldsymbol{\mu}_{\mathbf{Z}})$$

$$\mathbf{K}_{\mathbf{XX}}=\frac{1}{2}\Re\left(\mathbf{K}_{\mathbf{Z}}+\tilde{\mathbf{K}}_{\mathbf{Z}}\right)$$
$$\mathbf{K}_{\mathbf{XY}}=\frac{1}{2}\Im\left(\tilde{\mathbf{K}}_{\mathbf{Z}}-\mathbf{K}_{\mathbf{Z}}\right)$$

then we can write

$$f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})=f_{\Re(\mathbf{Z}),\Im(\mathbf{Z})}\left(\Re(\mathbf{z}),\Im(\mathbf{z})\right)=f_{\mathbf{Z}}(\mathbf{z}) \tag{4.12}$$

Thus $f_{\mathbf{Z}}(\mathbf{z})$ is the PDF of the complex Gaussian random vector $\mathbf{Z}$.
Therefore, the PDF of a complex Gaussian random vector $\mathbf{Z}$ is the joint PDF of its real and imaginary parts $\mathbf{X}$ and $\mathbf{Y}$, respectively, $\mathbf{X}$ and $\mathbf{Y}$ being jointly Gaussian.

Joint CF of X and Y

$$\begin{aligned}
\varphi_{\mathbf{X},\mathbf{Y}}(j\omega_1,j\omega_2) &= \mathbb{E}\left[e^{j\left(\boldsymbol{\omega}_1^{\top}\mathbf{X}+\boldsymbol{\omega}_2^{\top}\mathbf{Y}\right)}\right] \\
&= \mathbb{E}\left[e^{j\begin{bmatrix}\boldsymbol{\omega}_1^{\top} & \boldsymbol{\omega}_2^{\top}\end{bmatrix}\begin{bmatrix}\mathbf{X}\\ \mathbf{Y}\end{bmatrix}}\right] \\
&= e^{j\left(\boldsymbol{\omega}_1^{\top}\boldsymbol{\mu}_{\mathbf{X}}+\boldsymbol{\omega}_2^{\top}\boldsymbol{\mu}_{\mathbf{Y}}\right)-\frac{1}{2}\left(\boldsymbol{\omega}_1^{\top}\mathbf{K}_{\mathbf{XX}}\boldsymbol{\omega}_1+\boldsymbol{\omega}_2^{\top}\mathbf{K}_{\mathbf{YY}}\boldsymbol{\omega}_2+2\boldsymbol{\omega}_1^{\top}\mathbf{K}_{\mathbf{XY}}\boldsymbol{\omega}_2\right)}
\end{aligned} \tag{4.13}$$

A complex random vector whose pseudo-covariance matrix is zero is called a circular random vector. (A circular random vector need not be Gaussian.)

If the complex Gaussian vector $\mathbf{Z} = \mathbf{X} + j\mathbf{Y}$ is circular then,

$$\tilde{\mathbf{K}}_{\mathbf{Z}} = \mathbf{0}_{L\times L} \tag{5.1}$$

implying

$$\mathbf{K}_{\mathbf{XX}} = \mathbf{K}_{\mathbf{YY}}, \quad \mathbf{K}_{\mathbf{XY}} = -\mathbf{K}_{\mathbf{XY}}^{\top} \tag{5.2}$$

i.e., $\mathbf{X}$ and $\mathbf{Y}$ have equal covariance matrices, and the cross-covariance matrix of $\mathbf{X}$ and $\mathbf{Y}$ is skew-symmetric. (A real-valued skew-symmetric matrix always has zeros on its diagonals.)

Thus, if $\mathbf{X}$ and $\mathbf{Y}$ are independent real Gaussian vectors with equal covariance matrices, then $\mathbf{K}_{\mathbf{XY}} = -\mathbf{K}_{\mathbf{XY}}^{\top}$ is also satisfied, and $\mathbf{Z} = \mathbf{X} + j\mathbf{Y}$ is circular.

Note

- In wireless communication, we are interested in complex random vectors that have the circular symmetry property: $\mathbf{Z}$ is circular symmetric if $e^{j\theta}\mathbf{Z}$ has the same distribution of $\mathbf{Z}$ for any θ.
- If Z is complex circular with zero mean, then $e^{j\theta}Z$ will have the same second moments as Z. This is where the term circular comes from: a rotation of this random variable in the complex plane does not change its second moment description.
- When $L = 1$, we have $Z = X + jY$
  with $\mathbf{K}_{XX} = \mathrm{Var}\,(X) = \sigma_X^2$
  $\mathbf{K}_{YY} = \mathrm{Var}\,(Y) = \sigma_Y^2$
  $\mathbf{K}_{XY} = \mathrm{Cov}\,(X,Y) = \rho_{XY}\sigma_X\sigma_Y$
  If $Z = X + jY$ is a circular complex Gaussian random variable then $\sigma_X^2 = \sigma_Y^2$ and $\rho_{XY} = 0$ implying that X and Y are uncorrelated, and therefore independent.
- For any $L\times 1$ complex random vector (not necessarily Gaussian), with the real and imaginary parts having equal covariance matrices,
  uncorrelated real and imaginary parts $\Rightarrow$ circularity for all $L \geq 1$.
  but, circularity $\Rightarrow$ uncorrelated real and imaginary parts for $L = 1$ only.

PDF:

$$\begin{aligned} f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) &= \frac{1}{(2\pi)^L |\mathbf{K}|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}\begin{bmatrix}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})^\top & (\mathbf{y}-\boldsymbol{\mu}_{\mathbf{Y}})^\top\end{bmatrix} \mathbf{K}^{-1} \begin{bmatrix}\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}} \\ \mathbf{y}-\boldsymbol{\mu}_{\mathbf{Y}}\end{bmatrix}\right\} \\ &= \frac{1}{\pi^L |\mathbf{K}_{\mathbf{Z}}|} \exp\left\{-(\mathbf{z}-\boldsymbol{\mu}_{\mathbf{Z}})^{\mathsf{H}} \mathbf{K}^{-1}(\mathbf{z}-\boldsymbol{\mu}_{\mathbf{Z}})\right\} \\ &= f_{\mathbf{Z}}(\mathbf{z}) \end{aligned} \tag{5.3}$$

Thus when $\mathbf{Z}$ is a circular complex Gaussian random vector, we have

$$\begin{gathered} f_{\mathbf{Z}}(\mathbf{z}) = \frac{1}{\pi^L |\mathbf{K}_{\mathbf{Z}}|} \exp\left\{-(\mathbf{z}-\boldsymbol{\mu}_{\mathbf{Z}})^{\mathsf{H}} \mathbf{K}^{-1}(\mathbf{z}-\boldsymbol{\mu}_{\mathbf{Z}})\right\} \quad \mathbf{z} \in \mathbb{C}^L \text{ or } \Re(\mathbf{z}), \Im(\mathbf{z}) \in \mathbb{R}^L \\ \mathbf{Z} \sim \mathcal{CN}(\boldsymbol{\mu}_{\mathbf{Z}}, \mathbf{K}_{\mathbf{Z}}) \end{gathered} \tag{5.4}$$

i.e., $\mathbf{Z}$ is a circular complex Gaussian random vector with mean vector $\boldsymbol{\mu}_{\mathbf{Z}}$ and covariance matrix $\mathbf{K}_{\mathbf{Z}}$.

Rician and Rayleigh Random Variables

Let $X_1 \sim \mathcal{N}(\mu_1, \sigma^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma^2)$, X_1 and X_2 are independent. Let $Z = X_1 + jX_2 = Re^{j\Theta}$.

$$\left.\begin{aligned} \text{Let} \quad R &= \sqrt{X_1^2 + X_2^2} \\ \Theta &= \tan^{-1}\left(\frac{X_2}{X_1}\right) \end{aligned}\right\} \quad \Rightarrow \quad \begin{aligned} X_1 &= R\cos\Theta \\ X_2 &= R\sin\Theta \end{aligned} \tag{5.5}$$

$$\underbrace{\mathbf{J}\left(\frac{X_1, X_2}{R, \Theta}\right)}_{\text{Jacobian Matrix}} = \frac{\partial (X_1, X_2)}{\partial (R, \Theta)} = \begin{vmatrix} \dfrac{\partial X_1}{\partial R} & \dfrac{\partial X_2}{\partial R} \\ \dfrac{\partial X_1}{\partial \Theta} & \dfrac{\partial X_2}{\partial \Theta} \end{vmatrix} = \begin{vmatrix} \cos\Theta & \sin\Theta \\ -R\sin\Theta & R\cos\Theta \end{vmatrix} \tag{5.6}$$

Joint PDF of X_1 and X_2,

$$\begin{aligned} f_{X_1,X_2}(x_1, x_2) &= \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{1}{2}\left[\frac{(x_1-\mu_1)^2 + (x_2-\mu_2)^2}{\sigma^2}\right]\right\} \\ &= f_{X_1}(x_1) \cdot f_{X_2}(x_2) \end{aligned} \tag{5.7}$$

Joint PDF of R and θ,

$$\begin{aligned} f_{R,\Theta}(r,\theta) &= \mathbf{J} f_{X_1,X_2}(x_1,x_2)\Big|_{\substack{x_1 = r\cos\theta \\ x_2 = r\sin\theta}} \\ &= \frac{r}{2\pi\sigma^2} \exp\left\{-\frac{1}{2}\left[\frac{(r\cos\theta-\mu_1)^2 + (r\sin\theta-\mu_2)^2}{\sigma^2}\right]\right\} \\ &= \frac{r}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2}\left(r^2\cos^2\theta + \mu_1^2 - 2r\mu_1\cos\theta + r^2\sin^2\theta + \mu_2^2 - 2r\mu_2\sin\theta\right)\right\} \\ &= \frac{r}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2}\left[r^2 + \mu_1^2 + \mu_2^2 - 2r(\mu_1\cos\theta + \mu_2\sin\theta)\right]\right\} \end{aligned} \tag{5.8}$$

Take,

$$\begin{aligned} 2r(\mu_1\cos\theta + \mu_2\sin\theta) &= 2r\sqrt{\mu_1^2+\mu_2^2}\left(\frac{\mu_1}{\sqrt{\mu_1^2+\mu_2^2}}\cos\theta + \frac{\mu_2}{\sqrt{\mu_1^2+\mu_2^2}}\sin\theta\right) \\ &= 2r\sqrt{\mu_1^2+\mu_2^2}\,(\cos\phi\cos\theta + \sin\phi\sin\theta) \\ &= 2r\sqrt{\mu_1^2+\mu_2^2}\cos(\theta-\phi) \end{aligned} \tag{5.9}$$

where

$$\cos\phi = \frac{\mu_1}{\sqrt{\mu_1^2+\mu_2^2}}, \quad \sin\phi = \frac{\mu_2}{\sqrt{\mu_1^2+\mu_2^2}} \quad \text{and} \quad \tan\phi = \frac{\mu_2}{\mu_1}$$

Using eqn. (5.9) in (5.8)

$$\begin{aligned} f_{R,\Theta}(r,\theta) &= \frac{r}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2}\left[r^2 + \mu_1^2 + \mu_2^2 - 2r\sqrt{\mu_1^2+\mu_2^2}\cos(\theta-\phi)\right]\right\} \\ &= \frac{r}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2}\left[r^2 + \mu_1^2 + \mu_2^2 - 2r\sqrt{\mu_1^2+\mu_2^2}\cos\left(\theta - \tan^{-1}\frac{\mu_2}{\mu_1}\right)\right]\right\} \\ & \qquad 0 \le r < \infty, \; -\pi \le \theta \le \pi \end{aligned} \tag{5.10}$$

If we want to find the PDF of R, then

$$\begin{aligned} f_R(r) &= \int_{-\pi}^{\pi} f_{R,\Theta}(r,\theta)\,d\theta \\ &= \frac{r}{\sigma^2}\exp\left\{-\frac{(r^2+\mu_1^2+\mu_2^2)}{2\sigma^2}\right\} \times \left\{\frac{1}{2\pi}\int_{-\pi}^{\pi}\exp\left\{\frac{r\sqrt{\mu_1^2+\mu_2^2}}{\sigma^2}\cos\left(\theta - \tan^{-1}\frac{\mu_2}{\mu_1}\right)\right\}d\theta\right\} \end{aligned}$$

we know that,

$$\underbrace{I_0(x)}_{\text{Modified Bessel function of zeroth order and first kind}} \triangleq \frac{1}{2\pi}\int_{-\pi}^{\pi}\exp\{x\cos(\theta+\phi)\}\,d\theta$$

$\therefore$ PDF of R,

$$f_R(r) = \frac{r}{\sigma^2}\exp\left\{-\frac{(r^2+\mu_1^2+\mu_2^2)}{2\sigma^2}\right\} I_0\left(\frac{r}{\sigma^2}\sqrt{\mu_1^2+\mu_2^2}\right), \quad r \ge 0 \tag{5.11}$$

$f_R(r)$ - marginal PDF : R has a Rician distribution when $\mu_1^2 + \mu_2^2 > 0$.

Note

- $X_1 \sim \mathcal{N}\left(\mu_1, \sigma^2\right), \quad X_2 \sim \mathcal{N}\left(\mu_2, \sigma^2\right)$
  - Different mean $\mu_1 \neq \mu_2 \neq 0$
  - Same variance
  - X_1 and X_2 are independent

  Then $R = \sqrt{X_1^2 + X_2^2}$ follows Rician distribution (non-zero mean).

- $X_1 \sim \mathcal{N}\left(0, \sigma^2\right), \quad X_2 \sim \mathcal{N}\left(0, \sigma^2\right)$
  - $\mu_1 = \mu_2 = 0$
  - Same variance
  - X_1 and X_2 are independent

  Then $R = \sqrt{X_1^2 + X_2^2}$ follows Rayleigh distribution (zero mean).
  Rayleigh PDF :

$$f_R(r) = \frac{r}{\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\}, \quad r \geq 0$$

$$\text{where} \quad I_0(0) = 1$$

- $X_1 + jX_2$: Complex Gaussian random variable.

$$R = \sqrt{X_1^2 + X_2^2} \quad \Theta = \tan^{-1}\left(\frac{X_2}{X_1}\right)$$

$$f_{X_1,X_2}(x_1, x_2) \to f_{R,\Theta}(r, \theta)$$

$$f_R(r) = \int_{-\pi}^{\pi} f_{R,\Theta}(r, \theta)\, d\theta$$

$$f_\Theta(\theta) = \int_0^\infty f_{R,\Theta}(r, \theta)\, dr$$

- We know that X_1 and X_2 are independent

$$\Rightarrow \quad f_{X_1,X_2}(x_1, x_2) = f_{X_1}(x_1)\, f_{X_2}(x_2)$$

$$\text{But,} \quad f_{R,\Theta}(r, \theta) \neq f_R(r)\, f_\Theta(\theta)$$

  R and Θ are not independent of each other for Rician distribution.

- If $\mu_1 = 0$ and $\mu_2 = 0$, we know that,

$$f_R(r) = \frac{r}{\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\}, \quad r \geq 0$$

$$\begin{aligned} f_\Theta(\theta) &= \int_0^\infty f_{R,\Theta}(r, \theta)\, dr \\ &= \int_0^\infty \frac{r}{2\pi\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\} dr \\ &= \frac{1}{2\pi} \int_0^\infty \frac{r}{\sigma^2} \exp\left\{-\frac{r^2}{2\sigma^2}\right\} dr \\ &= \frac{1}{2\pi} \int_0^\infty f_R(r)\, dr \\ &= \frac{1}{2\pi} \end{aligned} \tag{5.12}$$

  Θ is uniformly distributed over $[-\pi, \pi]$ with PDF :

$$f_\Theta(\theta) = \frac{1}{2\pi}$$

$\therefore$ For $\mu_1 = \mu_2 = 0$, we have,

$$\begin{aligned} f_{R,\Theta}(r,\theta) &= \frac{r}{2\pi\sigma^2}\exp\left\{-\frac{r^2}{2\sigma^2}\right\} \\ &= \frac{1}{2\pi}\cdot\frac{r}{\sigma^2}\exp\left\{-\frac{r^2}{2\sigma^2}\right\} \\ &= f_R(r)\cdot f_\Theta(\theta) \end{aligned}$$

$\Rightarrow$ R and Θ are independent

$$\begin{aligned} f_R(r) &: \text{ Rayleigh} \\ f_\Theta(\theta) &: \text{ Uniform} \end{aligned}$$

Note: X_1 and X_2 are independent
This does not hold when $\mu_1^2 + \mu_2^2 > 0$.

- The phase of a signal with a Rayleigh envelop is uniformly distributed over $[-\pi, \pi]$ and independent of the envelop.
- The phase of a signal with a Rician envelop is neither uniformly distributed over $[-\pi, \pi]$, nor independent of the envelop.
- To find $f_{R^2}(y)$:

$$F_{R^2}(y) = \Pr\left(R^2 \le y\right) = \Pr\left(R \le \sqrt{y}\right) = F_R\left(\sqrt{y}\right)$$

$$f_{R^2}(y) = \frac{f_R\left(\sqrt{y}\right)}{2\sqrt{y}}$$

We know that,

$$f_R(r) = \frac{r}{\sigma^2}\exp\left\{-\frac{\left(r^2+\mu_1^2+\mu_2^2\right)}{2\sigma^2}\right\} I_0\left(\frac{r}{\sigma^2}\sqrt{\mu_1^2+\mu_2^2}\right), \quad r \ge 0$$

$$\begin{aligned} f_{R^2}(y) &= \frac{1}{2\sqrt{y}}\cdot\frac{\sqrt{y}}{\sigma^2}\exp\left\{-\frac{\left(y+\mu_1^2+\mu_2^2\right)}{2\sigma^2}\right\} I_0\left(\frac{\sqrt{y\left(\mu_1^2+\mu_2^2\right)}}{\sigma^2}\right) \\ &= \frac{1}{2\sigma^2}\exp\left\{-\frac{\left(y+\mu_1^2+\mu_2^2\right)}{2\sigma^2}\right\} I_0\left(\frac{\sqrt{y\left(\mu_1^2+\mu_2^2\right)}}{\sigma^2}\right), \quad y \ge 0 \end{aligned} \tag{5.13}$$

R^2 has a non central chi-squared distribution with degrees of freedom $= 2$ $\left(\chi^2(2)\right)$ when $\mu_1^2+\mu_2^2 > 0$.

- When $\mu_1 = \mu_2 = 0$, we get,

$$f_{R^2}(y) = \frac{1}{2\sigma^2}\exp\left\{-\frac{y}{2\sigma^2}\right\}, \quad y \ge 0 \tag{5.14}$$

R^2 has a central chi-squared distribution with degrees of freedom $= 2$ $\left(\chi^2(2)\right)$ when $\mu_1 = \mu_2 = 0$.
Also called, R^2 has an exponential distribution when $\mu_1 = \mu_2 = 0$.

Advanced Topics in Communication Systems	Thurs 22 Jan 2026
Lecture 6	

Chi-Squared Distribution

Let $X \sim \mathcal{N}\left(\mu, \sigma^2\right)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad -\infty < x < \infty$$

X^2 has a chi-squared distribution with one degree of freedom. Denoted as $\chi^2(1)$.
When $\mu \neq 0$, X^2 has a non central $\chi^2(1)$ distribution.
When $\mu = 0$, X^2 has a central $\chi^2(1)$ distribution.
Let $Y = X^2$

$$\begin{aligned} F_Y(y) &= \Pr(Y \leq y) = \Pr\left(X^2 \leq y\right) = \Pr(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= \Pr(X \leq \sqrt{y}) - \Pr(X < -\sqrt{y}) \\ &= \Pr(X \leq \sqrt{y}) - \Pr(X \leq -\sqrt{y}) + \Pr(X = -\sqrt{y}) \end{aligned} \tag{6.1}$$

Differentiating both sides wrt y, we get,

$$f_Y(y) = \frac{1}{2\sqrt{y}} f_X(\sqrt{y}) + \frac{1}{2\sqrt{y}} f_X(-\sqrt{y}) + \frac{d}{dy}\Pr(X = -\sqrt{y}) \tag{6.2}$$

when $f_X(x)$ has no impulse, $\Pr\left(X = -\sqrt{y}\right) = 0$.

$$\begin{aligned} \therefore \quad f_Y(y) &= \frac{1}{2\sqrt{y}}\left[f_X(\sqrt{y}) + f_X(-\sqrt{y})\right] \\ &= \frac{1}{2\sqrt{2\pi\sigma^2 y}}\left[\exp\left\{-\frac{\left(\sqrt{y}-\mu\right)^2}{2\sigma^2}\right\} + \exp\left\{-\frac{\left(\sqrt{y}+\mu\right)^2}{2\sigma^2}\right\}\right] \end{aligned} \tag{6.3}$$

PDF of non central chi-squared distribution with degree of freedom $= 1$ $\left(\chi^2(1)\right)$.

Characteristic Function of Y

$$\begin{aligned} \varphi_Y(j\omega) &= \mathbb{E}\left[e^{j\omega Y}\right] = \mathbb{E}\left[e^{j\omega X^2}\right] \\ &= \int_{-\infty}^{\infty} e^{j\omega x^2} f_X(x)\, dx \\ &= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{j\omega x^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx \\ &= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{j\omega x^2} e^{-\frac{\left(x^2-2\mu x+\mu^2\right)}{2\sigma^2}}\, dx \end{aligned}$$

put $v = \frac{x}{\sigma} \Rightarrow dv = \frac{dx}{\sigma}$,

$$\begin{aligned} \varphi_Y(j\omega) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{j\omega\sigma^2 v^2} e^{-\frac{\left(v^2\sigma^2-2\mu\sigma v+\mu^2\right)}{2\sigma^2}}\, dv \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{j\omega\sigma^2 v^2} e^{-\frac{1}{2}\left(v^2-2\mu\frac{v}{\sigma}+\frac{\mu^2}{\sigma^2}\right)}\, dv \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[\left(1-2j\omega\sigma^2\right)v^2-2\mu\frac{v}{\sigma}+\frac{\mu^2}{\sigma^2}\right]}\, dv \end{aligned}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{\left(1-2j\omega\sigma^2\right)}{2}\left[v^2 - 2\mu \frac{v}{\sigma(1-2j\omega\sigma^2)} + \frac{\mu^2}{\sigma^2(1-2j\omega\sigma^2)}\right]} dv$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{\left(1-2j\omega\sigma^2\right)}{2}\left[v - \frac{\mu}{\sigma(1-2j\omega\sigma^2)}\right]^2} e^{-\frac{\left(1-2j\omega\sigma^2\right)}{2}\left[\frac{\mu^2}{\sigma^2(1-2j\omega\sigma^2)} - \frac{\mu^2}{\sigma^2(1-2j\omega\sigma^2)^2}\right]} dv$$

$$= e^{-\frac{\mu^2}{2\sigma^2}\left(1-\frac{1}{1-2j\omega\sigma^2}\right)} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{\left(1-2j\omega\sigma^2\right)}{2}\left(v - \frac{\mu}{\sigma(1-2j\omega\sigma^2)}\right)^2} dv$$

Now,

$$\frac{1}{\sqrt{2\pi A^2}} \int_{-\infty}^{\infty} e^{-\frac{(v-b)^2}{2A^2}} dv = 1$$

$$b = \frac{\mu}{\sigma\left(1-2j\omega\sigma^2\right)}, \quad A = \frac{1}{\left(1-2j\omega\sigma^2\right)^{\frac{1}{2}}}$$

$$\therefore \quad \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{\left(1-2j\omega\sigma^2\right)}{2}\left(v - \frac{\mu}{\sigma(1-2j\omega\sigma^2)}\right)^2} dv = \frac{1}{\left(1-2j\omega\sigma^2\right)^{\frac{1}{2}}}$$

$$\Rightarrow \quad \varphi_Y\left(j\omega\right) = \frac{e^{-\frac{\mu^2}{2\sigma^2}\frac{\left(-2j\omega\sigma^2\right)}{1-2j\omega\sigma^2}}}{\left(1-2j\omega\sigma^2\right)^{\frac{1}{2}}}$$

$$\left.\varphi_Y\left(j\omega\right) = \frac{e^{\frac{j\omega\mu^2}{1-2j\omega\sigma^2}}}{\left(1-2j\omega\sigma^2\right)^{\frac{1}{2}}}\right\} \quad \begin{array}{l}\text{Non central } \chi^2(1) \text{ CF}\\ \text{when } \mu \neq 0\end{array} \tag{6.4}$$

When $\mu = 0$, we get,

$$\left.\varphi_Y\left(j\omega\right) = \frac{1}{\left(1-2j\omega\sigma^2\right)^{\frac{1}{2}}}\right\} \quad \begin{array}{l}\text{Central } \chi^2(1) \text{ CF}\\ \text{when } \mu = 0\end{array} \tag{6.5}$$

PDF is,

$$\left.f_Y\left(y\right) = \frac{1}{\sqrt{2\pi\sigma^2 y}} e^{-\frac{y}{2\sigma^2}}, \quad y \geq 0\right\} \quad \text{Central } \chi^2(1) \text{ PDF} \tag{6.6}$$

When $X_1 \sim \mathcal{N}\left(\mu_1, \sigma^2\right)$, $X_2 \sim \mathcal{N}\left(\mu_2, \sigma^2\right)$, X_1 and X_2 are independent. Let $Z = X_1 + jX_2$ and $R^2 = X_1^2 + X_2^2$

$$\begin{aligned}\varphi_{R^2}\left(j\omega\right) &= \mathbb{E}\left[e^{j\omega\left(X_1^2+X_2^2\right)}\right]\\ &= \mathbb{E}\left[e^{j\omega X_1^2}\right]\mathbb{E}\left[e^{j\omega X_2^2}\right]\\ &= \frac{e^{\frac{j\omega\mu_1^2}{1-2j\omega\sigma^2}}}{\left(1-2j\omega\sigma^2\right)^{\frac{1}{2}}} \cdot \frac{e^{\frac{j\omega\mu_2^2}{1-2j\omega\sigma^2}}}{\left(1-2j\omega\sigma^2\right)^{\frac{1}{2}}}\end{aligned}$$

$$\left.\varphi_{R^2}\left(j\omega\right) = \frac{e^{\frac{j\omega\left(\mu_1^2+\mu_2^2\right)}{1-2j\omega\sigma^2}}}{\left(1-2j\omega\sigma^2\right)}\right\} \quad \begin{array}{l}\text{Non central } \chi^2(2) \text{ CF}\\ \text{when } \mu_1^2+\mu_2^2 > 0\end{array} \tag{6.7}$$

Recall,

$$f_{R^2}\left(y\right) = \frac{1}{2\sigma^2}\exp\left\{-\frac{\left(y+\mu_1^2+\mu_2^2\right)}{2\sigma^2}\right\} I_0\left(\frac{\sqrt{y\left(\mu_1^2+\mu_2^2\right)}}{\sigma^2}\right), \quad y \geq 0$$

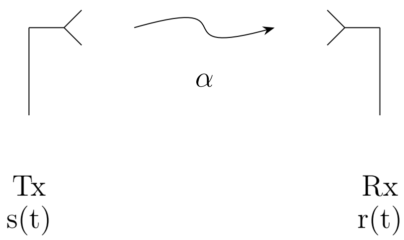
When $\mu_1 = \mu_2 = 0$,

$$\left.\varphi_{R^2}\left(j\omega\right) = \frac{1}{1-2j\omega\sigma^2}\right\} \quad \begin{array}{l}\text{Central } \chi^2(2) \text{ CF or Exponential CF}\\ \text{with mean } 2\sigma^2\end{array} \tag{6.8}$$

Recall,

$$f_{R^2}\left(y\right) = \frac{1}{2\sigma^2}\exp\left\{-\frac{y}{2\sigma^2}\right\}, \quad y \geq 0$$

Exponential Distribution



Complete baseband signal is,

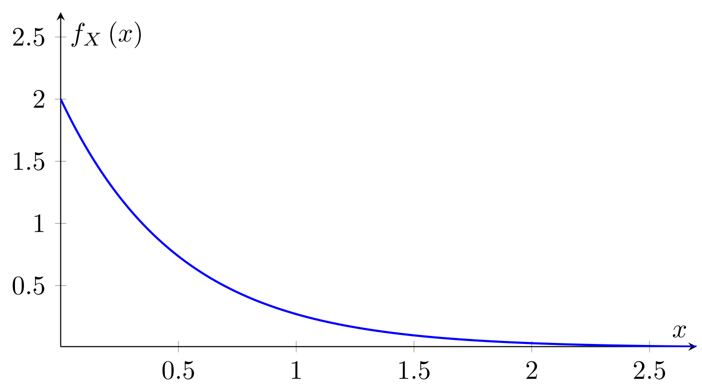
$$r(t) = \alpha e^{-j\theta} s(t) + n(t) \tag{7.1}$$

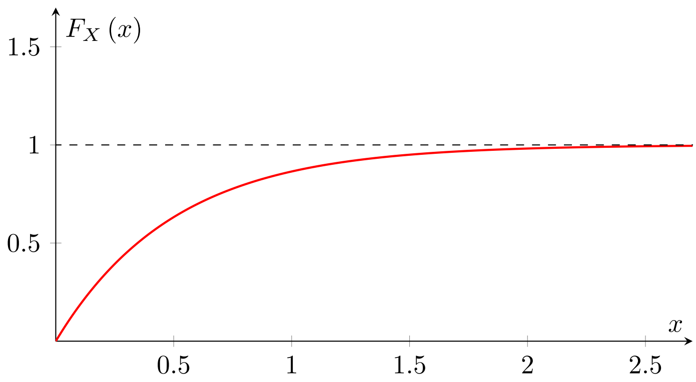
- $s(t)$: transmitted signal
- $n(t)$: AWGN
- α : Rayleigh
- θ : uniform
- α & θ : independent
- $\alpha e^{-j\theta}$: fading gain
- α^2 : exponentially distributed (envelope of signal received over a Rayleigh fading channel)

PDF of exponentially distributed RV X,

$$\begin{gathered} f_X(x) = \lambda e^{-\lambda x} = \frac{1}{2\sigma^2} \exp\left\{-\frac{x}{2\sigma^2}\right\}, \quad x \geq 0 \\ \lambda = \frac{1}{2\sigma^2} \end{gathered} \tag{7.2}$$

$$\mathbb{E}[X] = \frac{1}{\lambda} \tag{7.3}$$





Exponential CDF,

$$F_X(x) = \int_0^\infty f_X(x)\,dx = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \tag{7.4}$$

Exponential CF,

$$\varphi_X(h\omega) = \frac{1}{(1 - j\omega\lambda^{-1})}, \quad \text{where } \lambda = \frac{1}{2\sigma^2} \tag{7.5}$$

Note that X can be expressed as,

$$X = Y_1^2 + Y_2^2 \tag{7.6}$$

where Y_1, Y_2 are iid, $\mathcal{N}\left(0, \sigma^2\right) \equiv \mathcal{N}\left(0, \frac{1}{2\lambda}\right)$. $\sqrt{X}$ is Rayleigh distributed.

Gamma Distribution

- Widely used distribution.
- Continuous probability distribution.
- Its importance is largely due to its relation to exponential distribution, Erlang distribution and chi-squared distribution.
- $\left.\begin{array}{l}\text{Exponential distribution} \\ \text{Erlang distribution} \\ \text{Chi-squared distribution}\end{array}\right\}$ Special cases of the Gamma distribution

A continuous random variable X is said to be have a Gamma distribution with parameters $\alpha > 0$ and $\beta > 0$, if its PDF is given by,

$$f_X(x) = \frac{x^{\alpha-1} e^{-\frac{x}{\beta}}}{\Gamma(\alpha)\beta^\alpha}, \quad x \geq 0,\ \alpha > 0 \text{ and } \beta > 0 \tag{7.7}$$

$$\begin{aligned} \text{Mean } \mathbb{E}[X] &= \int_0^\infty x f_X(x)\,dx \\ &= \int_0^\infty \frac{x \cdot x^{\alpha-1} e^{-\frac{x}{\beta}}}{\Gamma(\alpha)\beta^\alpha} dx \\ &= \int_0^\infty \frac{x^\alpha e^{-\frac{x}{\beta}}}{\Gamma(\alpha)\beta^\alpha} dx \end{aligned}$$

put $v = \frac{x}{\beta} \Rightarrow dv = \frac{dx}{\beta}$,

$$
\begin{aligned}
\mathbb{E}\left[X\right] &= \int_0^\infty \frac{v^\alpha \beta^\alpha e^{-v}}{\Gamma(\alpha)\beta^\alpha}\beta dv \\
&= \frac{\beta}{\Gamma(\alpha)}\int_0^\infty v^\alpha e^{-v} dv \\
&= \frac{\beta}{\Gamma(\alpha)}\Gamma(\alpha+1) \\
&= \frac{\beta}{\Gamma(\alpha)}\alpha\Gamma(\alpha) \\
&= \alpha\beta
\end{aligned} \tag{7.8}
$$

$$
\mathrm{Var}\left(X\right) = \alpha\beta^2 \tag{7.9}
$$

Gamma CF,

$$
\varphi_X\left(j\omega\right) = \frac{1}{\left(1 - j\omega\beta\right)^\alpha} \tag{7.10}
$$

Special Cases

$$
f_X\left(x\right) = \frac{x^{\alpha-1}e^{-\frac{x}{\beta}}}{\Gamma(\alpha)\beta^\alpha}
$$

1. $\alpha = 1$

$$
\begin{aligned}
\underbrace{f_X\left(x\right)}_{\text{Exponential}} &= \frac{1}{\beta}e^{-\frac{x}{\beta}} = \lambda e^{-\lambda x}, \quad x > 0 \\
\lambda &= \frac{1}{\beta}
\end{aligned} \tag{7.11}
$$

2. $\alpha = n$, n is a positive integer, and $\frac{1}{\beta} = \lambda$, we get,

$$
\begin{aligned}
f_X\left(x\right) &= \frac{x^{n-1}e^{-\frac{x}{\beta}}}{\Gamma(n)\beta^n} \\
&= \frac{\lambda^n x^{n-1}e^{-\lambda x}}{(n-1)!}, \quad x \geq 0
\end{aligned} \tag{7.12}
$$

$$
\mathbb{E}\left[X\right] = \frac{n}{\lambda} \tag{7.13}
$$

$$
\varphi_X\left(j\omega\right) = \frac{1}{\left(1 - j\omega\lambda^{-1}\right)^n} \tag{7.14}
$$

- If we sum n independent exponential random variables, then we will get a Gamma random variable. We can express X as,

$$
X = Y_1^2 + Y_2^2 + \cdots + Y_{2n}^2 \tag{7.15}
$$

  $Y_1, Y_2, \ldots, Y_{2n}$ are iid $\mathcal{N}\left(0, \frac{1}{2\lambda}\right)$.

- When $\alpha = n$, where n is a positive integer, and $\frac{1}{\beta} = \lambda$

$$
F_X\left(x\right) = \begin{cases} 1 - \displaystyle\sum_{k=0}^{n-1} \frac{\left(\lambda x\right)^k}{k!}e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \tag{7.16}
$$

- X has a central chi-squared $\chi^2(2n)$ distribution with CF as,

$$
\varphi_X\left(j\omega\right) = \left[\frac{1}{\left(1 - 2j\omega(2\lambda)^{-1}\right)^{\frac{1}{2}}}\right]^{2n} \tag{7.17}
$$

  $n = 1 \Rightarrow$ Exponential distribution

Chi-Squared distribution with n-degrees of freedom

Let $X = Y_1^2 + Y_2^2 + \cdots + Y_n^2 = \sum_{i=1}^{n} Y_i^2$; $Y_1, Y_2, \ldots, Y_n$ iid $\mathcal{N}\left(0, \frac{1}{\lambda}\right)$. Then X is said to be centrally chi-squared distributed with n degrees of freedom, or said to have a central $\chi^2(n)$ distribution.

$$\text{CF } \varphi_X(j\omega) = \frac{1}{\left(1 - 2j\omega\lambda^{-1}\right)^{\frac{n}{2}}} \tag{7.18}$$

Gamma distribution with $\alpha = \frac{n}{2}$, $\beta = \frac{2}{\lambda}$

$$\therefore \quad \text{PDF } f_X(x) = \frac{\lambda^{\frac{n}{2}} x^{\frac{n}{2}-1}}{2^{\frac{n}{2}}\Gamma\left(\frac{n}{2}\right)} e^{-\frac{\lambda x}{2}}, \quad x \geq 0 \tag{7.19}$$

$$\text{Mean } \mathbb{E}[X] = \frac{n}{\lambda} \tag{7.20}$$

Putting $\lambda^{-1} = \sigma^2$, we get,

$$\text{PDF } f_X(x) = \frac{x^{\frac{n}{2}-1} e^{-\frac{x}{2\sigma^2}}}{\left(2\sigma^2\right)^{\frac{n}{2}}\Gamma\left(\frac{n}{2}\right)}, \quad x \geq 0 \tag{7.21}$$

$$\text{CF } \varphi_X(j\omega) = \frac{1}{\left(1 - 2j\omega\sigma^2\right)^{\frac{n}{2}}} \tag{7.22}$$

$$\text{Mean } \mathbb{E}[X] = n\sigma^2 \tag{7.23}$$

Thus, $X = Y_1^2 + Y_2^2 + \cdots + Y_n^2$; $Y_1, Y_2 \ldots, Y_n$ iid $\mathcal{N}\left(0, \sigma^2\right)$.

$$n = 1, \quad \text{central } \chi^2(1) \Rightarrow \varphi_X(j\omega) = \frac{1}{\left(1 - 2j\omega\sigma^2\right)^{\frac{1}{2}}} \tag{7.24}$$

$$n = 2, \quad \text{central } \chi^2(2) \text{ or exponential} \Rightarrow \varphi_X(j\omega) = \frac{1}{\left(1 - 2j\omega\sigma^2\right)} \tag{7.25}$$

The noncentral chi-squared (χ^2) random variable

Let $X = Y_1^2 + Y_2^2 + \cdots + Y_n^2 = \sum_{i=1}^{n} Y_i^2$. The noncentral χ^2 random variable with n degrees of freedom is defined similarly to a central χ^2 random variable in which Y_i's are independent Gaussians with common variance σ^2 but with different means denoted by μ_i.
The PDF of X is of the form,

$$f_X(x) = \frac{1}{2\sigma^2}\left(\frac{x}{s^2}\right)^{\frac{n-2}{4}} e^{-\left(\frac{s^2+x}{2\sigma^2}\right)} I_{\frac{n}{2}-1}\left(\frac{s}{\sigma^2}\sqrt{x}\right), \quad x > 0 \tag{7.26}$$

where s is defined as,

$$s = \sqrt{\sum_{i=1}^{n} \mu_i^2} \tag{7.27}$$

$$I_\alpha(x) = \text{Modified bessel function of the first kind and order } \alpha$$

The function $I_0(x)$ can be written as,

$$\begin{aligned} I_0(x) &= \frac{1}{\pi}\int_0^{\pi} e^{\pm x\cos\phi} d\phi \\ &= \frac{1}{2\pi}\int_0^{2\pi} e^{x\cos\phi} d\phi \\ &= \sum_{k=0}^{\infty}\left(\frac{x^k}{2^k k!}\right)^2 \end{aligned} \tag{7.28}$$

For $x > 1$ can be approximated by,

$$I_0(x) \approx \frac{e^x}{\sqrt{2\pi x}} \tag{7.29}$$

In general,

$$I_\alpha(x) = \sum_0^\infty \frac{\left(\frac{x}{2}\right)^{\alpha+2k}}{k!\Gamma(\alpha+k+1)}, \quad x \geq 0 \tag{7.30}$$

The CDF of X, when $n = 2m$, can be written in the form,

$$F_X(x) = \begin{cases} 1 - Q_m\left(\frac{s}{\sigma}, \frac{\sqrt{x}}{\sigma}\right) & x > 0 \\ 0 & \text{otherwise} \end{cases} \tag{7.31}$$

where $Q_m(a, b)$ is the generalized Marcum Q-function and is defined as,

$$Q_m(a, b) = \int_b^\infty x\left(\frac{x}{a}\right)^{m-1} e^{-\frac{x^2+a^2}{2}} I_{m-1}(ax)dx \tag{7.32}$$

$$\text{CF } \varphi_X(j\omega) = \left(\frac{1}{1-2j\omega\sigma^2}\right)^{\frac{n}{2}} e^{\frac{j\omega s^2}{1-2j\omega\sigma^2}} \tag{7.33}$$

Nakagami-m Distribution

Both the Rayleigh distribution and the Rice distribution are frequently used to describe the statistical fluctuations of signals received from a multipath fading channel.

Anothr distribution that is frequently used to characterize the statistics of signals transmitted through multipath fading channels is the Nakagami-m distribution.

- Related to Gamma distribution
- Used to model physical phenomena, such as, communications engineering, multimedia etc.

We know the PDF of Gamma distributed random variable X,

$$f_X(x) = \frac{x^{\alpha-1}e^{-\frac{x}{\beta}}}{\Gamma(\alpha)\beta^\alpha}$$

Putting $\alpha = m$, $\alpha\beta = \Omega$, we get,

$$f_X(x) = \frac{m^m x^{m-1} e^{-\frac{mx}{\Omega}}}{\Gamma(m)\Omega^m}, \quad x \geq 0 \tag{7.34}$$

$$\varphi_X(j\omega) = \frac{1}{\left(1 - j\omega\frac{\Omega}{m}\right)^m} \tag{7.35}$$

$$\mathbb{E}[X] = \Omega \tag{7.36}$$

- X has a Gamma distribution with parameter m and mean Ω
- $\sqrt{X}$ has a Nakagami-m distribution

Note

In other words, if Y is Nakagami distributed then Y^2 is Gamma distributed.

If X has a Gamma distribution with parameter m and mean Ω, $Y = \sqrt{X}$ has a Nakagami-m distribution,

$$\begin{aligned} F_Y(y) &= \Pr(Y \leq y) = \Pr\left(\sqrt{X} \leq y\right) = \Pr\left(X \leq y^2\right) \\ &= F_X\left(y^2\right) \end{aligned} \tag{7.37}$$

$$\therefore \quad f_Y(y) = 2y f_X\left(y^2\right) \tag{7.38}$$

$$\because \quad f_X(x) = f_X(x) = \frac{m^m x^{m-1} e^{-\frac{mx}{\Omega}}}{\Gamma(m)\Omega^m}, \quad x \geq 0$$

PDF of Nakagami,

$$f_Y(y) = \frac{2m^m y^{2m-1} e^{-\frac{my^2}{\Omega}}}{\Gamma(m)\Omega^m}, \quad y \geq 0, m > 0 \text{ and } \Omega > 0 \tag{7.39}$$

Mean square value,

$$\mathbb{E}\left[Y^2\right] = \Omega \tag{7.40}$$

$$\left.\begin{aligned} \mathbb{E}[X] &= \Omega \\ \mathbb{E}\left[Y^2\right] &= \Omega \end{aligned}\right\} \Rightarrow \begin{array}{l}\text{Mean value} \\ \text{of Gamma}\end{array} = \begin{array}{l}\text{Mean square} \\ \text{value of} \\ \text{Nakagami}\end{array}$$

Nakagami distribution with $m = 1$ is the Rayleigh distribution,

$$f_Y(y) = \frac{2y}{\Omega} e^{-\frac{y^2}{\Omega}}, \quad y \geq 0 \tag{7.41}$$

- In communication systensm the signal amplitude values of a signal received over a fading channel can be modeled as a Rayleigh distribution.
- Compared to the Rayleigh distribution, the Nakagami distribution gives greater flexibilty to model random fluctuations (fading) channels in communications.

$$m = \frac{1}{2}, \quad f_Y(y) = \frac{2}{\sqrt{2\pi\Omega}} e^{-\frac{y^2}{2\Omega}}, \quad y \geq 0 \quad \text{One-sided Gaussian fading} \tag{7.42}$$

$$m = 1, \quad f_Y(y) = \frac{2y}{\Omega} e^{-\frac{y^2}{\Omega}}, \quad y \geq 0 \quad \text{Rayleigh fading} \tag{7.43}$$

$$m = \infty, \quad f_Y(y) = \delta\left(y - \sqrt{\Omega}\right), \quad \text{Deterministic, no fading} \tag{7.44}$$

Received signal (complex baseband) over a fading channel,

$$r(t) = Y e^{-j\theta} s(t) + \underbrace{n(t)}_{\text{AWGN}} \tag{7.45}$$

- Y : fading amplitude or envelope
- θ : fading phase

As m increases, the amount of fading decreases. The worst kind of fading is generally one-sided Gaussian fading.

Hence, we choose $m \geq \frac{1}{2}$ for a Nakagami fading channel.

$$\text{CF } \varphi_{Y^2}(j\omega) = \varphi_X(j\omega) = \frac{1}{\left(1 - j\omega\frac{\Omega}{m}\right)^m} \tag{7.46}$$

$$\begin{aligned} &\text{As } m \to \infty, \quad \varphi_X(j\omega) \to \frac{1}{e^{-j\omega\Omega}} = e^{j\omega\Omega} \\ &\Rightarrow \quad f_X(x) = \delta(x - \Omega) \quad \text{when } m \to \infty \\ \therefore \quad & f_Y(y) = f_{\sqrt{X}}(y) = \delta\left(y - \sqrt{\Omega}\right) \quad \text{when } m \to \infty \end{aligned} \tag{7.47}$$